Self Join in

A **Self Join** is a type of join in where a table is joined with itself. This allows you to compare rows within the same table. It's useful for situations where the data has a hierarchical or relational structure within a single table, such as employees and their managers, or in cases where you need to compare values in different rows of the same table.

Self Join Example

Example Scenario: Booth Relationships
Let's assume that booths in the farmers market have relationships where one booth could be a "parent booth" for another booth. We want to find out which booths are linked together based on booth type.

Query:

```
SELECT A.booth_number AS Booth1, B.booth_number AS RelatedBooth
FROM booth A
JOIN booth B ON A.booth_type = B.booth_type
WHERE A.booth_number <> B.booth_number;
```

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1. Basic: Find Vendors Operating in the Same Booth Type

Query:

```
SELECT A.vendor_id AS Vendor1, B.vendor_id AS Vendor2
FROM vendor_booth_assignments A
JOIN vendor_booth_assignments B ON A.booth_number = B.booth_number
WHERE A.vendor_id <> B.vendor_id;
```

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2. Intermediate: Find Vendors that Share Booths on the Same Market Date

```
SELECT A.vendor_id AS Vendor1, B.vendor_id AS Vendor2, A.market_date
FROM vendor_booth_assignments A
JOIN vendor_booth_assignments B ON A.booth_number = B.booth_number AND A.market_date = B.market_date
WHERE A.vendor_id <> B.vendor_id;
```

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3. Advanced: Identify Products with the Same Category Sold by Different Vendors

```
SELECT P1.product_name AS Product1, P2.product_name AS Product2, C.product_category_name
FROM product P1
JOIN product P2 ON P1.product_category_id = P2.product_category_id
JOIN product_category C ON P1.product_category_id = C.product_category_id
WHERE P1.product_id <> P2.product_id;
```

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4. Basic: List Customers from the Same Zip Code

```
SELECT A.customer_first_name AS Customer1, B.customer_first_name AS Customer2,
A.customer_zip
FROM customer A
JOIN customer B ON A.customer_zip = B.customer_zip
WHERE A.customer_id <> B.customer_id;
```

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5. Intermediate: Customers Buying Products from the Same Vendor

```
SELECT A.customer_id AS Customer1, B.customer_id AS Customer2, A.vendor_id, A.product_id
FROM customer_purchases A
JOIN customer_purchases B ON A.vendor_id = B.vendor_id AND A.product_id = B.product_id
WHERE A.customer_id <> B.customer_id;
```